EXPERIMENT - 6

AIM

To demonstrate session tracking using HttpSession in a Servlet-based Online Bus Pass Application System by implementing a login module that authenticates users and maintains their session during usage.

ALGORITHM:

1. Input:

- *The user enters their username and password in the bus pass login form.*
- *The servlet validates credentials and creates a session if login is successful.*

2. Steps for Implementing Session Tracking

Step 1: Create the Login Form (HTML)

- *Design a login page for the Online Bus Pass system.*
- *Include input fields for:*
  - *Username*
  - *Password*
- *Add a submit button to send data to the login servlet.*

Step 2: Create the Login Servlet

- *Develop a servlet class to handle login requests.*
- *Retrieve username and password from the request.*
- *Authenticate using predefined values (or database).*
- *Create an HttpSession object if credentials are valid.*

Step 3: Validate User Credentials

- *Compare entered details with stored values.*
- *If valid:*
    - *Create session*
    - *Store username using session.setAttribute("username", username)*
- *If invalid:*
    - *Display error message or redirect to login page*

Step 4: Create and Track User Session

- *Use HttpSession to store user details.*
- *Maintain session across pages of the bus pass system.*
- *Optionally set session timeout using setMaxInactiveInterval().*

Step 5: Redirect to Bus Pass Dashboard

- *After successful login, redirect user to:*
    - *Bus Pass Home Page / Dashboard*
- *Check session attribute before allowing access.*

Step 6: Logout and Session Invalidation

- *Provide logout option in the system.*

- *On logout:*
    - *Call session.invalidate()*
    - *Redirect user to login page*

Step 7: Session Expiration Handling

- *If session expires or user logs out:*
    - *Restrict access to protected pages*
    - *Redirect back to login page*

CODE:

```
<!DOCTYPE html>
<html>
<head>
<title>Bus Pass Login</title>

<style>
body{
    margin:0;
    font-family: Arial;
    background: linear-gradient(135deg,#1e3c72,#2a5298);
    color:white;
}

header{
    background:#1e88e5;
    padding:15px;
```

```
    text-align:center;
    font-size:24px;
}

.container{
    width:350px;
    margin:100px auto;
    padding:25px;
    background: rgba(255,255,255,0.15);
    border-radius:12px;
    text-align:center;
    box-shadow:0 0 15px rgba(0,0,0,0.3);
}

input{
    width:100%;
    padding:10px;
    margin:10px 0;
    border:none;
    border-radius:5px;
}

button{
    width:100%;
    padding:10px;
    border:none;
    border-radius:25px;
```

```
    background: linear-gradient(45deg,#00e5ff,#00c853);
    color:white;
    cursor:pointer;
}

button:hover{
    transform:scale(1.05);
}
</style>

</head>

<body>

<header>Online Bus Pass Application System</header>

<div class="container">
<h2>Login</h2>

<form action="LoginServlet" method="post">
<input type="text" name="username" placeholder="Enter Username" required>
<input type="password" name="password" placeholder="Enter Password" required>
<button type="submit">Login</button>
</form>

</div>
```

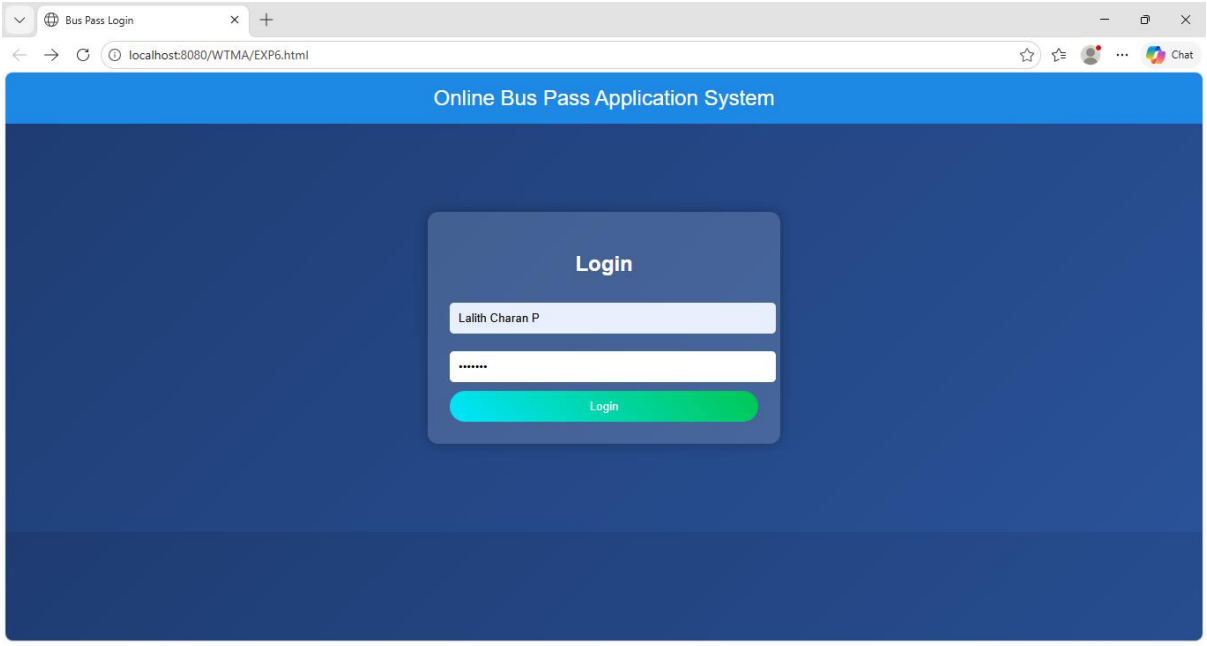
```
</body>
</html>
package WTMA;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;
import java.io.*;

@WebServlet("/DashboardServlet")
public class DashboardServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if(session == null || session.getAttribute("username") == null){
            response.sendRedirect("EXP6.html");
            return;
        }

        String user = (String) session.getAttribute("username");

        response.setContentType("text/html");
        PrintWriter pw = response.getWriter();
```

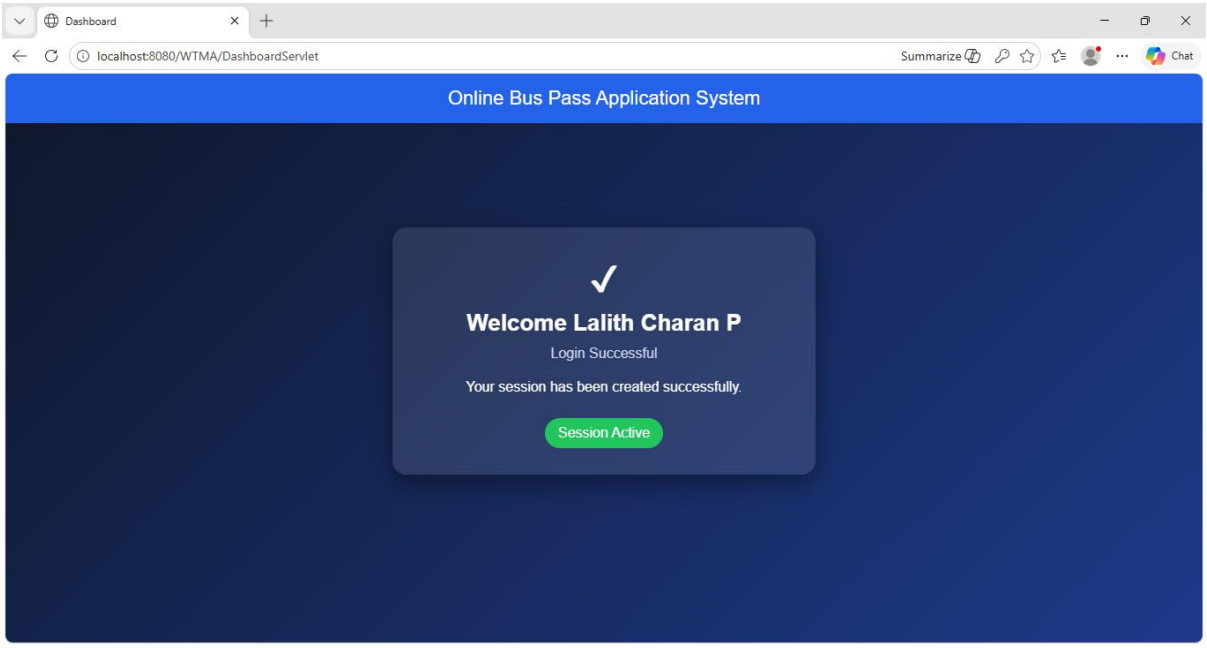
```
pw.println("<html><head><title>Dashboard</title>");

pw.println("<style>");
pw.println("body{margin:0;font-family:Arial;background:linear-gradient(135deg,#0f172a,#1e3a8a);color:white;}");
pw.println("header{background:#2563eb;padding:15px;text-align:center;font-size:22px;}");
pw.println(".container{display:flex;justify-content:center;align-items:center;height:80vh;}");
pw.println(".card{width:420px;padding:30px;border-radius:16px;background:rgba(255,255,255,0.1);backdrop-filter:blur(10px);box-shadow:0 10px 30px rgba(0,0,0,0.4);text-align:center;}");
pw.println(".title{font-size:26px;font-weight:bold;margin-bottom:10px;}");
pw.println(".subtitle{font-size:16px;color:#cbd5f5;margin-bottom:20px;}");
pw.println(".badge{display:inline-block;padding:8px 15px;border-radius:20px;background:#22c55e;color:white;margin-top:10px;}");
pw.println(".icon{font-size:40px;margin-bottom:10px;}");
pw.println("</style>");

pw.println("</head><body>");

pw.println("<header>Online Bus Pass Application System</header>");

pw.println("<div class='container'>");
pw.println("<div class='card'>");
```

```
        pw.println("<div class='icon'>✓</div>");
        pw.println("<div class='title'>Welcome " + user + "</div>");
        pw.println("<div class='subtitle'>Login Successful</div>");

        pw.println("<p>Your session has been created successfully.</p>");
        pw.println("<div class='badge'>Session Active</div>");

        pw.println("</div>");
        pw.println("</div>");

        pw.println("</body></html>");
    }
}
package WTMA;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;
import java.io.*;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
```

```
        String password = request.getParameter("password");
        if(username.equals("Lalith Charan P") &&
password.equals("Lalith@")){

            HttpSession session = request.getSession();
            session.setAttribute("username", username);

            response.sendRedirect(request.getContextPath() +
"/DashboardServlet");

        } else {
            response.sendRedirect("EXP6.html");
        }
    }
}
```

OUTPUT:





RESULT:

*The session tracking mechanism using **HttpSession** in a Servlet-based web application was successfully implemented. The user was able to log in using valid credentials, and a session was created to store user-specific data.*